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In [13]:

```
import numpy as np
import matplotlib.pyplot as plt
import tkinter as tk
from tkinter import filedialog
from statsmodels.tsa.seasonal import seasonal_decompose
root = tk.Tk()
root.withdraw()
root.attributes("-topmost", True)

file_path = filedialog.askopenfilename(
    title="Select AirPassengers.csv",
    filetypes=[("CSV files", "*.csv")]
)

root.destroy()
df.head()
```

Out[13]:

	Month	#Passengers
0	1949-01	112
1	1949-02	118
2	1949-03	132
3	1949-04	129
4	1949-05	121

In [9]:

```
data = pd.read_csv(file_path)

# Convert Month to datetime

data['Month'] = pd.to_datetime(data['Month'])

# Set Month as index
data.set_index('Month', inplace=True)

# Regular differencing
data['passengers_diff'] = (
    data['#Passengers'] - data['#Passengers'].shift(1)
)

# Seasonal adjustment
result = seasonal_decompose(
    data['#Passengers'], model='additive', period=12
)
data['passengers_sea_diff'] = result.resid

# Log transformation
data['passengers_log'] = np.log(data['#Passengers'])

# Log transformation and regular differencing
data['passengers_log_diff'] = (
    data['passengers_log'] - data['passengers_log'].shift(1)
)

# Log transformation, regular differencing and seasonal adjustment
result = seasonal_decompose(
    data['passengers_log_diff'].dropna(),
    model='additive',
    period=12
)
data['passengers_log_seasonal_diff'] = result.resid

# Plot all six graphs
plt.figure(figsize=(16, 16))

plt.subplot(6, 1, 1)
plt.plot(data['#Passengers'], label='Original')
plt.legend(loc='best')
plt.title('Original Data')
plt.xlabel('Year')
plt.ylabel('No of passengers')

plt.subplot(6, 1, 2)
plt.plot(data['passengers_diff'], label='Regular Difference')
plt.legend(loc='best')
plt.title('Regular Differencing')
plt.xlabel('Year')
plt.ylabel('Differenced No of passengers')

plt.subplot(6, 1, 3)
plt.plot(data['passengers_sea_diff'], label='Seasonal Adjustment')
plt.legend(loc='best')
plt.title('Seasonal Adjustment')
```

```
plt.xlabel('Year')
plt.ylabel('Seasonally adjusted passengers')

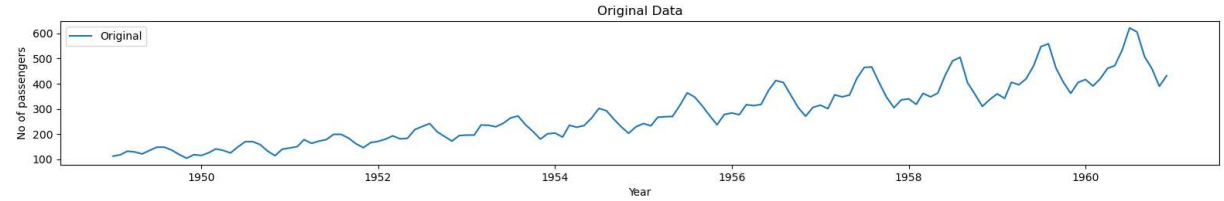
plt.subplot(6, 1, 4)
plt.plot(data['passengers_log'], label='Log Transformation')
plt.legend(loc='best')
plt.title('Log Transformation')
plt.xlabel('Year')
plt.ylabel('Log(No of passengers)')

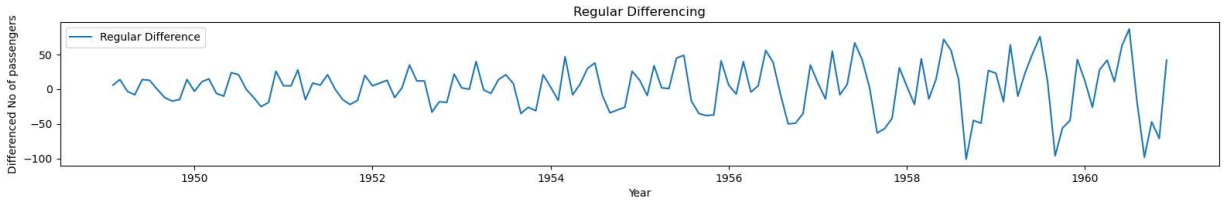
plt.subplot(6, 1, 5)
plt.plot(
    data['passengers_log_diff'],
    label='Log Transformation and Regular Differencing'
)
plt.legend(loc='best')
plt.title('Log Transformation and Regular Differencing')
plt.xlabel('Year')
plt.ylabel('Diff(Log(No of passengers))')

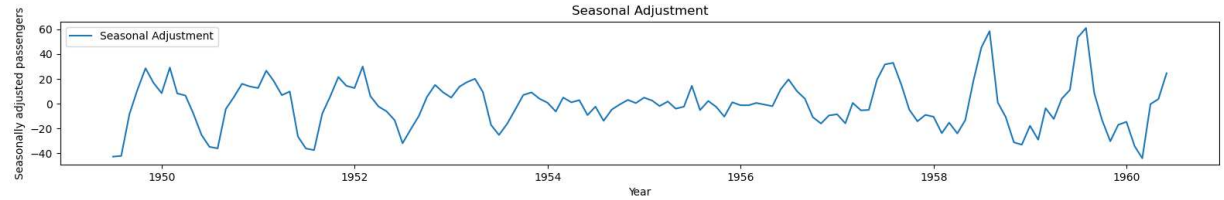
plt.subplot(6, 1, 6)
plt.plot(
    data['passengers_log_seasonal_diff'],
    label='Log Transformation and Seasonal Adjustment'
)
plt.legend(loc='best')
plt.title(
    'Log Transformation, Regular Differencing and Seasonal Adjustment'
)
plt.xlabel('Year')
plt.ylabel('Adjusted differenced log passengers')

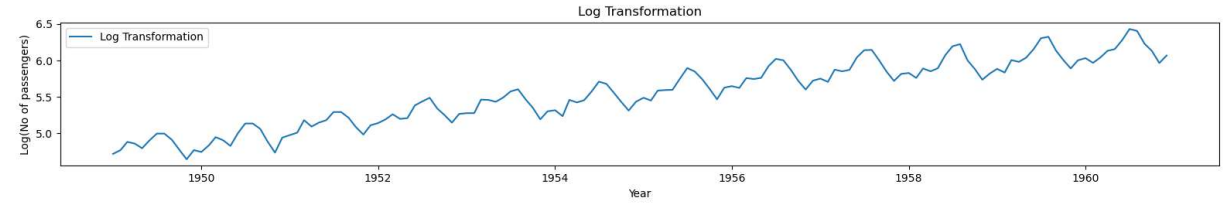
plt.tight_layout()
plt.show()

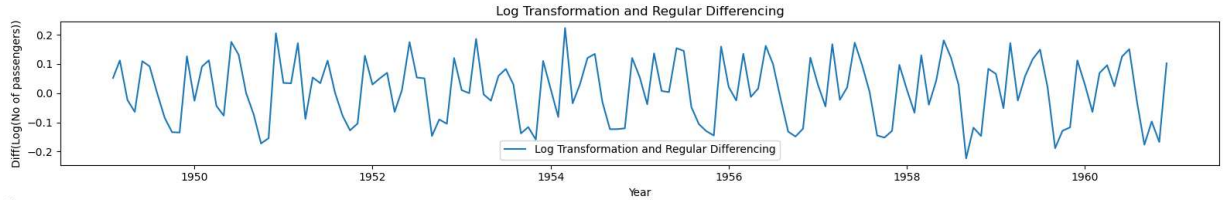
# Overall view
data.plot(kind='line', figsize=(12, 6))
plt.title('Overall Time Series Comparison')
plt.show()
```

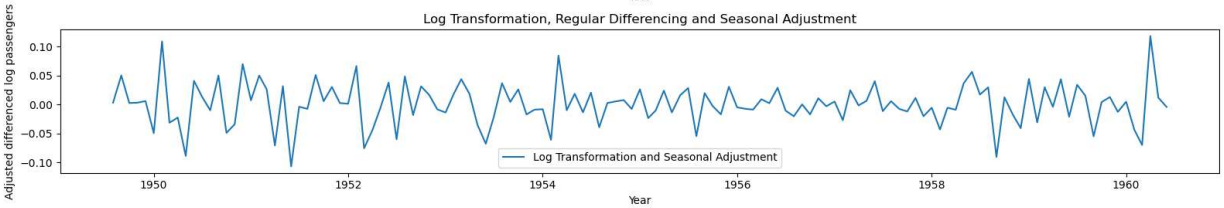

Original Data
Original
No of passengers
Year

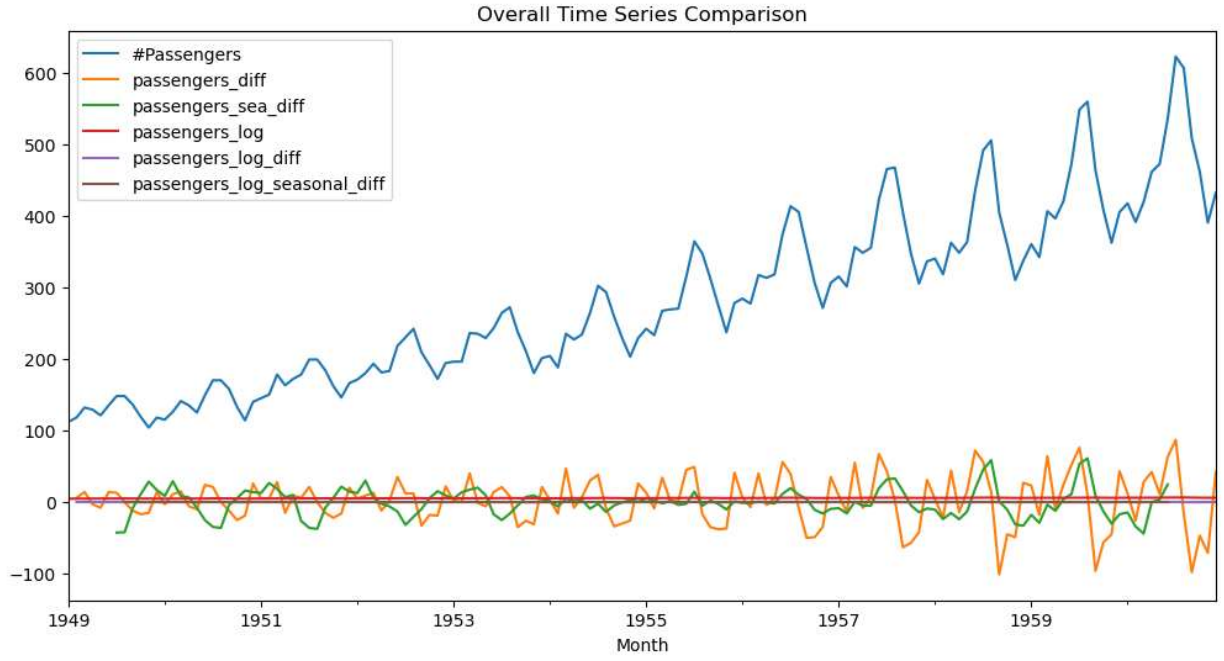

Regular Differencing
Regular Difference
Differenced No of passengers
Year


Seasonal Adjustment
Seasonal Adjustment
Seasonally adjusted passengers
Year


Log Transformation
Log Transformation
Log(No of passengers)
Year


Log Transformation and Regular Differencing
Log Transformation and Regular Differencing
Diff(Log(No of passengers))
Year


Log Transformation, Regular Differencing and Seasonal Adjustment
Log Transformation and Seasonal Adjustment
Adjusted differenced log passengers
Year


Overall Time Series Comparison
#Passengers
passengers_diff
passengers_sea_diff
passengers_log
passengers_log_diff
passengers_log_seasonal_diff
Month